

1

Source Data

LLMBar Natural +
Adversarial

n = 100

n = 100

2

Counterfactual Generation

Qwen3-32B
style rewriter

formal ↔ casual

100 cf.

3

NLI Content Verification

DeBERTa-v3-large

threshold ≥ 0.90

88 pass

4

Dual Pairwise Scoring

LLM Judge
(A/B swap)

2 trials per pair

173 comp.

5

Statistical Analysis

BT model + GEE
decomposition

+ Likert comparison